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Intrauterine Mepivacaine Instillation vs Placebo for Pain During IUD Placement

A Randomized Clinical Trial

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IMPORTANCE Concern about pain during intrauterine device (IUD) placement is common and reduces selection of an IUD. Reducing procedure-related pain may support informed contraceptive decision-making and patient autonomy and has also been associated with improved patient satisfaction and procedural experience.

OBJECTIVE To determine whether intrauterine instillation of mepivacaine reduces pain during IUD placement in nulliparous individuals.

DESIGN, SETTING, AND PARTICIPANTS Multicenter, double-blind, randomized, placebo-controlled trial conducted at 11 outpatient gynecology, maternal health, and youth clinics in Sweden from May 2021 through June 2024. Nulliparous individuals aged 18 to 31 years opting for IUDs with smaller-diameter placement tubes (levonorgestrel 19.5 mg, 13.5 mg, or copper IUD) were eligible. Key exclusion criteria included ongoing pregnancy, IUD replacement, pelvic infection, and allergy to local anesthetics. A total of 370 participants were randomized.

INTERVENTIONS Participants were randomly assigned (1:1) (185 per group) to intrauterine instillation of 10 mL of mepivacaine (20 mg/mL) or sodium chloride (9 mg/mL) via a hydrososonography catheter 2 minutes before IUD placement. Participants, clinicians, and outcome assessors were blinded to treatment allocation.

MAIN OUTCOMES AND MEASURES The primary outcome was clinician-adjusted difference in pain scores during IUD placement, measured on a 100-mm visual analog scale. The primary analysis included all randomized participants with available primary outcome data. Prespecified secondary analyses included tolerability and overall experience.

RESULTS A total of 370 participants were randomized. Mean age was 21.7 (SD, 3.0) years in the mepivacaine group and 21.7 (SD, 2.9) years in the placebo group. Mean pain score during IUD placement was 43.8 mm (SD, 24.7) in the mepivacaine group and 58.6 mm (SD, 21.5) in the placebo group (mean difference, 14.8 mm [95% CI, 10.0-19.6]; $P < .001$). Adjustment for individual clinician yielded similar results (mean difference, 14.2 mm [95% CI, 9.7-18.6]; $P < .001$). Tolerable pain was reported by 172 of 175 participants (98.3%) in the mepivacaine group and 159 of 174 (91.4%) in the placebo group (relative risk, 1.08 [95% CI, 1.02-1.13]; NNT, 15 [95% CI, 9-43]).

CONCLUSIONS AND RELEVANCE Intrauterine instillation of 10 mL of mepivacaine (20 mg/mL) significantly reduced pain and improved tolerability during IUD placement compared with instillation of placebo saline.

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Intrauterine devices (IUDs) are an effective long-acting reversible contraceptive method with high user satisfaction.^{1,2} Concern about pain during IUD placement is common and affects contraceptive decision-making.³⁻⁵

Effective evidence-based options for pain management during IUD placement are limited; therefore, no specific method is consistently recommended in clinical guidelines.⁶⁻⁸ Existing approaches during IUD placement (including intra-cervical/intrauterine lidocaine gel, intracervical injections, and paracervical block) have demonstrated reduction in pain in some randomized trials. However, these methods may cause substantial discomfort at administration, potentially reducing acceptability.^{6,9-13} In addition, injectable local anesthetic techniques are not routinely available in all outpatient settings where IUDs are placed. Consequently, effective, minimally invasive, and tolerable approaches to pain relief remain an important unmet clinical need.

Previous studies on intrauterine instillation of local anesthetics have generally shown limited effectiveness for pain relief.^{14,15} Intrauterine instillation has previously been investigated as pain relief for endometrial biopsy and hysteroscopy with pain-reducing effects.¹⁶⁻¹⁸ In a recent randomized clinical trial among participants receiving 52-mg levonorgestrel-releasing IUDs, intrauterine instillation of mepivacaine was associated with lower pain scores and higher satisfaction compared with placebo.¹⁹ The present study applied the same protocol to IUDs with smaller-diameter placement tubes. Although IUDs with smaller-diameter placement tubes are generally associated with lower pain scores than larger devices, pain during placement remains common and may still be moderate to severe for some individuals.²⁰ The primary objective of this study was to determine the pain-relieving effect of intrauterine instillation of 10 mL of mepivacaine (20 mg/mL) during placement of IUDs with smaller-diameter placement tubes in nulliparous individuals.

Methods

Study Design and Setting

This multicenter, double-blind, randomized, placebo-controlled trial was conducted across 11 outpatient gynecology, maternal health, and youth clinics in Sweden. The methods, including the standardized IUD placement procedure, have been described.¹⁹ The study is reported in accordance with the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline. The study protocol is available in [Supplement 1](#) and the statistical analysis plan in [Supplement 2](#).

Recruitment and IUD placement were performed by 28 experienced midwives and gynecologists (eTable 1 in [Supplement 3](#) lists trial sites and clinicians per site). All clinics received a presentation on hydrosonography catheter use. No practical training was provided. Mepivacaine was selected because it has a faster onset and less potential toxicity compared with lidocaine.²¹ The IUD types (placement tube diameter) included a copper IUD (Nova-T 380, 3.6 mm) and the 19.5-mg or 13.5-mg levonorgestrel-releasing hormonal IUD (3.8 mm). Study IUD arms do not protrude outside the placement tube, and all were produced by Bayer AG.

Key Points

Question Does intrauterine instillation of 10 mL of mepivacaine (20 mg/mL) reduce pain during intrauterine device (IUD) placement in nulliparous individuals?

Findings In this randomized clinical trial including 370 participants, mean pain scores during IUD placement were 43.8 in the mepivacaine group and 58.6 in the placebo group (adjusted difference, 14.2) on a 100-mm visual analog scale.

Meaning Intrauterine mepivacaine instillation reduced pain during IUD placement and could represent a simple approach to improve patient experience.

Participants were placed in the lithotomy position and a self-holding vaginal speculum was introduced. A 1.6-mm sterile hydrosonography catheter (Probimed) (image available in eFigure 1 in [Supplement 3](#)) was passed through the internal cervical os and judged to be in the uterine cavity if the thicker part of the catheter was well within the cervix (approximately 1-2 cm). The randomly assigned study fluid was instilled into the uterine cavity. If the catheter was unable to pass through the internal cervical os, the study drug was administered at the point the catheter could reach and documented in the case report form. After 2 minutes, a single-tooth tenaculum was applied to the cervix, uterine length measured with a uterine sound, and the IUD was placed according to the manufacturer's instructions. If the IUD could not be passed through the internal cervical os without further mechanical dilation, the placement attempt was considered unsuccessful.

Ethics

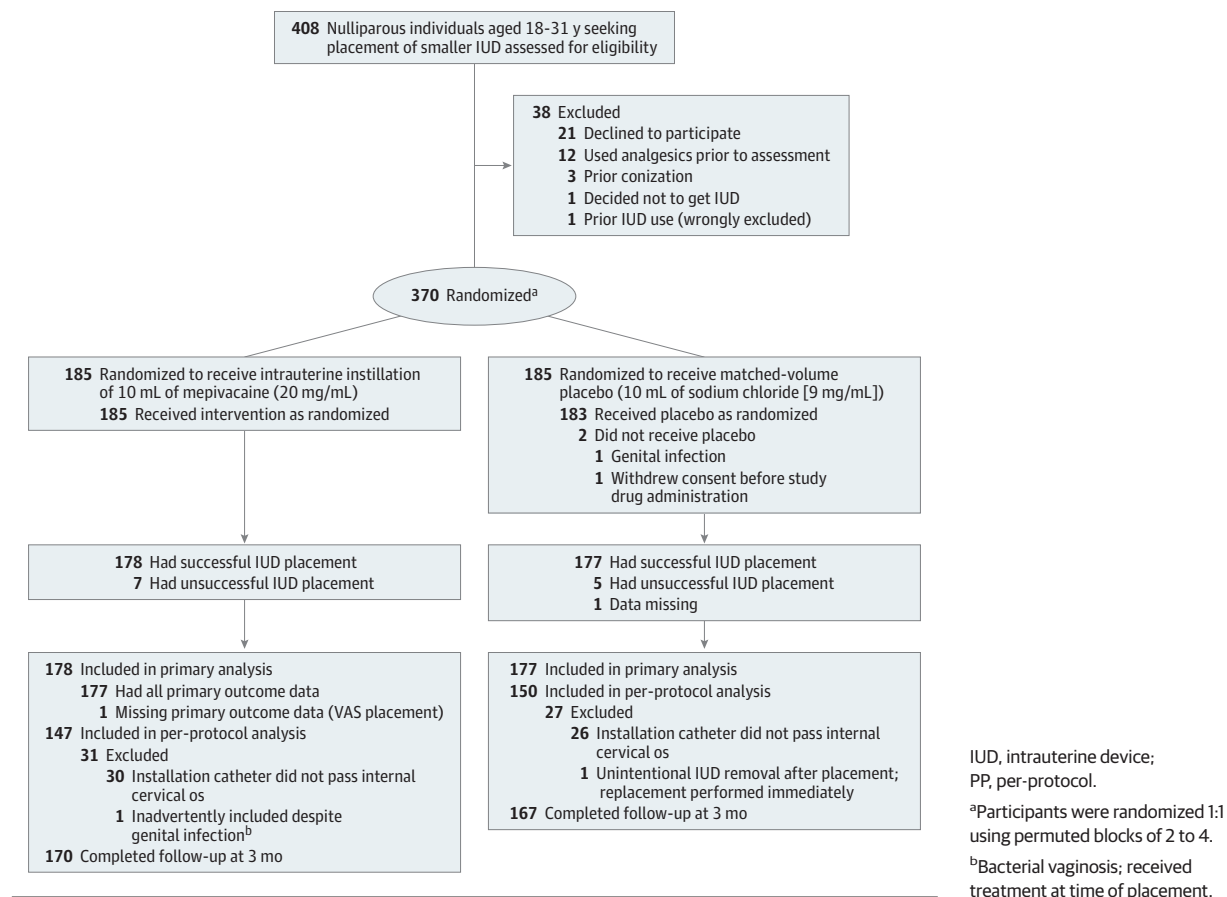
The study was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants after receipt of oral and written information about the study, including the randomization process, the possibility of receiving placebo, and the option to decline participation and discuss alternative pain management strategies, with the explicit mention of paracervical block.

Two methodologically identical studies (larger- and smaller-diameter IUDs) were registered with a single protocol and received a single registration number with the Swedish Ethical Review Authority (permit number 2021-00618, with amendments for including new study centers 2021-03087, 2021-04411) and the Swedish Medical Products Agency (EudraCT number 2020-002271-36). No changes were made to the protocol after submission. An independent monitor oversaw the study, and statistical analyses were overseen by medical statisticians.

Participants

Eligible participants were nulliparous individuals aged 18 to 31 years opting for an IUD for contraception. The upper age cutoff was chosen to represent the demographic group most likely to use an IUD to postpone first childbirth in Sweden, where the mean age at first birth at the time of the study was 29 years. Participants were eligible for IUD placement at any time during

Figure. Flow Diagram of a Randomized Clinical Trial of Intrauterine Mepivacaine for IUD Placement



the menstrual cycle, provided that a negative urine pregnancy test result was obtained before the procedure.

Additional eligibility criteria included ability to understand and provide informed consent after receiving both oral and written study information, proficiency in Swedish or English, agreement to randomization, and completion of required study questionnaires. Exclusion criteria included coming for IUD replacement, history of cervical conization, diagnosed cervical stenosis, body weight less than 40 kg, signs of genital infection, known uterine abnormalities, or conditions associated with increased risk of bleeding. Additional exclusion criteria were unexplained abnormal uterine bleeding, contraindications to local anesthetics, and use of other analgesic methods, including oral analgesics, prior to the procedure.

Randomization and Blinding

Randomization sequences were generated using Sealed Envelope software (seed 276798947588734) with 1:1 allocation in permuted blocks of 2 to 4. Randomization was performed by a nurse, nurse midwife, or physician not involved in the IUD placement procedure by sequential opening of sealed opaque envelopes containing the allocation code. Syringes containing either 10 mL of mepivacaine (20 mg/mL) or 10 mL of sodium chloride (9 mg/mL, placebo) were marked with the participation number and delivered on a tray at a designated spot

without meeting the clinician placing the IUD. The syringe fluids were indistinguishable in viscosity, color, and odor, thus maintaining blinding integrity for participants and clinicians. Instillation of placebo saline was used to maintain the double-blinded design, although it is not used in clinical practice for IUD placement. The randomization code key was concealed until completion of data collection and analysis of the primary outcome.

Procedures

Pain scores were assessed using a paper printed 100-mm visual analog scale (VAS), anchored at 0 mm as no pain and 100 mm as worst pain imaginable. Participants marked pain intensity score by drawing a vertical line on the VAS at different points; estimated worst menstrual cramping and preprocedural current pain (baseline), immediately after intrauterine instillation, at IUD placement, and 10 minutes after speculum removal. These points were chosen according to results of our pilot trial.²² At each point, participants also reported whether pain was tolerable (yes/no). Ten minutes after speculum removal, participants rated their overall experience with the placement procedure (easier than expected, as expected, or worse than expected). Participants were also asked whether they would like additional oral analgesia (yes/no), which was then administered if requested. Additional queries (yes/no)

Table 1. Participant Baseline Characteristics

Characteristic	Mepivacaine (n = 185)	Placebo (n = 183) ^a
Age, mean (SD), y	21.7 (3.0)	21.7 (2.9)
Reproductive history No./total (%)		
Previous pregnancy	13/184 (7.1)	17/182 (9.3)
Previous medical abortion	10/183 (5.4)	15/182 (8.2)
Previous surgical abortion	0/183	0/182
Previous spontaneous abortion	2/183 (1.1)	3/182 (1.6)
Menstruation information		
Last MP, mean (SD), d ^b	16.2 (9.1)	15.6 (11.1)
Amenorrhea, No. (%) ^c	30 (16.2)	20 (10.9)
Uncertain last MP, No. (%) ^b	1 (0.5)	4 (2.2)
VAS worst menstrual pain, mean (SD)	38.0 (26.0)	42.6 (23.6)
Previous IUD use, No. (%)	17 (9.2)	16 (8.7)
Severe dysmenorrhea (VAS ≥70 mm), No.	28 (15.1)	24 (13.1)
IUD type, No. (%)		
Levonorgestrel		
19.5 mg	147 (79.5)	147 (80.3)
13.5 mg	8 (4.3)	12 (6.6)
Copper	30 (16.2)	24 (13.1)

Abbreviations: IUD, intrauterine device; MP, menstrual period; VAS, visual analog scale.

^a One hundred eighty-five participants included but 2 were excluded due to infection and withdrawal.

^b First day of last menstrual period.

^c Amenorrhea due to current hormonal contraception.

included whether they would recommend the pain relief method to a friend and whether they would still have chosen an IUD had they known in advance how the procedure would feel. Study data were collected and managed using the Research Electronic Data Capture (REDCap) software hosted by Karolinska Institutet.²³

Outcomes

The primary outcome was the clinician-adjusted difference in VAS pain scores between the mepivacaine and placebo groups at the time of IUD placement. Secondary outcomes included pain at other time points, pain tolerability, and acceptability of the intervention.

Safety assessments included monitoring for adverse events at the placement visit and at a 3-month follow-up (see eMethods in Supplement 3).

Statistical Analysis

Data were analyzed using IBM SPSS Statistics, version 28.0.1. Because the primary outcome could only be assessed among participants who underwent a successful IUD placement, a full intention-to-treat analysis was not possible. The primary analysis therefore included all randomized participants with available primary outcome data. A per-protocol analysis was also performed and was restricted to participants with successful IUD placement in whom the catheter successfully passed the internal cervical os and the study drug was administered as intended. Unsuccessful placements are reported. No imputa-

tion of missing data was performed. Statistical significance was set at $P < .05$ (2-tailed).

VAS pain scores between groups were compared using independent t tests assuming unequal variances. Because individual clinicians may differ in procedural technique, communication, and approach to patient care, which could influence pain experienced during IUD placement, generalized linear mixed models were used to assess the effect of the intervention, with adjustment for individual clinician as a random effect (primary outcome). Secondary outcomes and associated relative risks (RRs) with 95% confidence intervals were analyzed and displayed graphically using R, version 4.4.2 (R Foundation). There was no data monitoring committee for this trial.

The sample size calculation was based on a pilot study with 86 randomized participants reporting a mean VAS pain score of 51 mm (SD, 28 mm) in the placebo group for IUDs with smaller placement tubes.²² A 20% reduction in the mean VAS pain score, corresponding to at least 10 mm, was considered clinically significant. To achieve 90% power with an $\alpha = .05$, the mepivacaine and placebo group required 174 participants each. To account for unsuccessful placements, enrollment targets were set at 185 participants per group.

Dysmenorrhea is associated with higher pain during IUD placement.²⁴ Post hoc exploratory subgroup analyses were conducted among participants with severe dysmenorrhea, defined as a VAS score 70 mm or greater at menstruation. This cutoff was based on prior literature but was not prespecified in the study protocol.^{25,26} These analyses aimed to assess the intervention effect among participants reporting severe dysmenorrhea, using generalized mixed models with the clinician as a random effect. In the subgroup analyses, between-group differences and 95% CIs were calculated using t tests, generalized linear mixed models, χ^2 tests, and Fisher exact tests, as appropriate. Since these were post hoc exploratory analyses, interaction testing was not performed and CIs were not adjusted for multiplicity; therefore, results should be interpreted with caution, and the findings should not be used for hypothesis testing.

Patient and Public Involvement

In developing this study protocol, we used insights from patients and a pilot study conducted between 2013 and 2017 and published in 2019.²² At that time, patient and public involvement was not mandatory; however, participants and clinicians found the method acceptable and easy to use. No patient organizations or laypeople were directly involved in planning the current trial, but recent publications mapping the social media landscape support that women experience IUD placement as painful and that clinicians can be dismissive of experienced pain.^{27,28}

Results

From May 2021 through June 2024, we assessed 408 individuals. A total of 370 participants were enrolled and randomized to intervention ($n = 185$) or placebo ($n = 185$). Details of enrollment are shown in the Figure. Twelve IUD placements

Table 2. Primary, Secondary, and Exploratory Outcomes According to Treatment Allocation

Outcome	Mepivacaine	Placebo	Effect estimate (95% CI) ^a	P value
Baseline measure	n = 185	n = 183		
Baseline pain, VAS score, mean (SD), mm	1.5 (4.8)	2.1 (5.7)	MD: 0.6 (0-1.7)	
Primary analysis	n = 177	n = 177		
Pain at IUD placement, VAS score, mean (SD), mm ^b	43.8 (24.7)	58.6 (21.5)	MD: 14.8 (10.0-19.6)	<.001
Adjusted difference ^c			MD: 14.2 (9.7-18.6)	<.001
Secondary analysis^d	n = 184	n = 182		
Instillation of study drug or placebo, VAS score, mean (SD), mm [No.]	18.2 (20.4) [184]	25.1 (23.2) [181]	MD: 6.9 (2.4-11.4)	
Ten minutes after placement, VAS score, mean (SD), mm [No.]	25.5 (23.0) [178]	34.2 (25.0) [177]	MD: 8.7 (3.7-13.8)	
Instillation pain tolerable, No. (%)	181/183 (98.9)	180 (98.9)	RR: 1.00 (0.98-1.02)	
Placement pain tolerable, No. (%)	172/175 (98.3)	159/174 (91.4)	RR: 1.08 (1.02-1.13)	
Easier than expected, No. (%)	93/176 (52.8)	59/172 (34.3)	RR: 1.54 (1.20-1.98)	
As expected, No. (%)	56/176 (31.8)	55/172 (32.0)	RR: 1.00 (0.73-1.35)	
Worse than expected, No. (%)	27/176 (15.3)	58/172 (33.7)	RR: 0.46 (0.30-0.68)	
Require additional pain medication 10 min after placement, No. (%)	92/176 (52.3)	115/177 (65.0)	RR: 0.78 (0.66-0.93)	
Would choose IUD for contraception again, No. (%)	164/176 (93.2)	141/172 (82.0)	RR: 1.14 (1.05-1.23)	
Would recommend the method for pain relief, No. (%)	139/176 (79.0)	118/172 (68.6)	RR: 1.15 (1.01-1.31)	
Vasovagal reaction, No. (%)	14/178 (7.9)	32/176 (18.2)	RR: 0.43 (0.24-0.78)	
Per-protocol analysis	n = 147	n = 150		
Instillation of study drug or placebo, VAS score, mean (SD), mm	20.1 (21.4)	28.2 (24.0)	MD: 8.1 (3.0-13.3)	
Pain at IUD placement, VAS score, mean (SD), mm ^b	42.2 (24.7)	59.1 (22.3)	MD: 16.9 (11.5-22.2)	<.001
Adjusted difference ^c			MD: 16.2 (11.2-21.2)	<.001
Ten minutes after placement, VAS score, mean (SD), mm	24.4 (22.1)	35.8 (25.5)	MD: 11.4 (5.9-16.8)	
Exploratory outcomes	n = 177	n = 177		
Severe IUD placement pain (VAS ≥70 mm), No. (%)	31 (17.5)	61 (34.5)	RR: 0.51 (0.35-0.74)	

Abbreviations: IUD, intrauterine device; MD, mean difference; RR, relative risk; VAS, visual analog scale.

^a Estimates are presented as mean differences for continuous variables and as relative risks for dichotomous outcomes.

^b From t test.

^c Generalized linear mixed models with clinician as random effect.

^d Percentages may not sum to 100 because of rounding.

were unsuccessful: 7 of 185 (3.8%) in the mepivacaine group and 5 of 183 (2.7%) in the placebo group. Participants and reasons for exclusion from the per-protocol analysis are presented in eTable 2 in Supplement 3. Table 1 presents the baseline characteristics, with no between-group differences in the proportions of IUD types chosen.

Results of the primary analysis and per-protocol analysis are presented in Table 2. In the primary analysis the mean VAS pain score during IUD placement was lower in the mepivacaine group (43.8 mm [SD, 24.7]) compared with the placebo group (58.6 mm [SD, 21.5]), with an absolute mean difference of 14.8 mm (95% CI, 10.0-19.6; $P < .001$). In the generalized linear mixed model, with clinician included as a random effect (primary outcome), the adjusted mean pain difference was 14.2 mm (95% CI, 9.7-18.6; $P < .001$), corresponding to a 23.7% decrease in pain (Table 2). During instillation, mean VAS pain scores were 18.2 mm (SD, 20.4) and 25.1 mm (SD, 23.2) in the mepivacaine and placebo group, respectively. In total, instillation pain was rated as tolerable by 361 of 365 participants (98.9%).

A higher proportion of participants in the mepivacaine group reported the placement pain as tolerable compared with the placebo group (172/175 [98.3%] vs 159/174 [91.4%]; RR, 1.08 [95% CI, 1.02-1.13]). The number needed to treat to achieve 1

additional participant reporting tolerable pain was 15 (95% CI, 9-43). Similarly, a greater proportion indicated that they would choose an IUD for contraception again (164/176 [93.2%] vs 141/172 [82.0%]; RR, 1.14 [95% CI, 1.05-1.23]). Fewer participants in the mepivacaine group rated the placement experience as worse than expected (27/176 [15.3%] vs 58/172 [33.7%]; RR, 0.46 [95% CI, 0.30-0.68]).

During IUD placement, fewer participants in the mepivacaine group than in the placebo group reported severe pain, defined as a VAS score 70 mm or greater (31/177 [17.5%] vs 61/177 [34.5%]; RR, 0.51 [95% CI, 0.35-0.74]).

No serious adverse events occurred. Vasovagal reactions were less common in the mepivacaine group (14/178 [7.9%]) than in the placebo group (32/176 [18.2%]; RR, 0.43 [95% CI, 0.24-0.78]). Additionally, 139 of 176 (79.0%) in the mepivacaine group would recommend the method for pain relief to a friend, compared with 118 of 172 (68.6%) in the placebo group (RR, 1.15 [95% CI, 1.01-1.31]). The adverse events at 3-month follow-up are summarized in Table 3. None of the adverse events reported at the 3-month follow-up reached a statistically significant difference between the groups.

In a post hoc subgroups analysis, 51 of 355 (14.4%) with successful IUD placements reported severe dysmenorrhea at

Table 3. Adverse Events at 3-Month Follow-Up

	No. (%)	
	Mepivacaine (n = 178)	Placebo (n = 177)
Worth the placement pain, No./total (%)	148/167 (88.6)	140/170 (82.4)
IUD expulsion, No./total (%)	5/168 (3.0)	2/170 (1.2)
Self-reported infection, No./total (%)	6/167 (3.6)	3/170 (1.8)
Urinary tract infection	1/167 (0.6)	0
Bacterial vaginosis	1/167 (0.6)	0
Candida	2/167 (1.2)	0
Nonspecified infection	2/167 (1.2)	3/170 (1.7)
Participants reporting visits to health care services, No./total (%) ^a	26/167 (15.6)	24/170 (14.1)
Reason for visit, No. (%)		
Pain	6 (3.4)	13 (7.3)
Bleeding	5 (2.8)	2 (1.1)
Suspected infection	0	1 (<1)
Confirm position/exclude partial or full expulsion of the IUD	2 (1.1)	3 (1.7)
Bacterial vaginosis	3 (1.7)	2 (1.1)
Candida	2 (1.1)	2 (1.1)
Pyelonephritis	1 (0.6)	0
Other nonsevere	8 (4.5)	5 (2.8)

Abbreviation: IUD, intrauterine device.

^a Participants could report more than 1 reason.

baseline (mepivacaine, 27/178 [15.2%]; placebo, 24/177 [13.6%]). There was a group difference in the mean VAS pain score during IUD placement mepivacaine, (57.4 mm [SD, 27.4]; placebo, 74.6 mm [SD, 20.7]). After adjustment for individual clinician, the absolute mean difference was 17.2 mm (95% CI, 3.4-31.0; $P = .02$), corresponding to a 23.1% decrease in pain (eTable 3 in Supplement 3).

Discussion

In this large, multicenter, double-blind, randomized, placebo-controlled trial, intrauterine instillation of 10 mL of mepivacaine (20 mg/mL) was associated with significantly reduced pain scores for nulliparous participants undergoing IUD placement using smaller-diameter placement tubes. Nearly all participants rated instillation pain as tolerable. Participants reported improved tolerability with mepivacaine instillation, supporting the acceptability of the intervention. Considering the frequency of IUD placements worldwide, even modest reductions in the proportion of individuals experiencing intolerable pain may have important clinical implications.

The findings of the present study are consistent with those of our previous randomized trial evaluating the same intervention during placement of a 52-mg hormonal IUD with a 4.4-mm-diameter placement tube.¹⁹ In that study, the mean clinician-adjusted reduction in pain was 12.2 mm, with a decrease from 67.4 mm to 55.2 mm. Thus, in both studies, intrauterine mepivacaine instillation reduced pain during IUD placement and other patient-reported outcomes, suggesting

that its beneficial effects are reproducible and generalizable beyond a specific IUD type or placement system.

Subjective outcomes such as pain are highly susceptible to placebo effects.²⁹ Previous studies of pain management during IUD placement have also primarily focused on pain scores, whereas patient-centered outcomes—such as tolerability, satisfaction, and willingness to choose an IUD again—have been reported less frequently. Therefore, current clinical guidelines promote a patient-centered approach rather than recommending a firsthand method for pain relief with IUD placement.⁶

This study was designed to assess the effect of mepivacaine compared with placebo, rather than the effect of intrauterine instillation per se. Because both groups underwent intrauterine instillation, any effect attributable to the instillation procedure would be expected to be balanced between groups. The possibility that the instillation itself influenced pain perception cannot be entirely excluded, although this is considered unlikely given the low pain scores observed during placebo administration. Inclusion of a third study group would have substantially increased the required sample size during a study period that was partially affected by the COVID-19 pandemic.

Exploratory post hoc analyses in the subset of participants with severe dysmenorrhea (VAS score ≥ 70 mm at menstruation) suggests that mepivacaine reduces pain and improves tolerability in this subgroup. However, because the study was neither designed nor powered to evaluate treatment effects for this group, these findings should be interpreted with caution.

Key strengths that increase internal validity and generalizability include the randomized, double-blind design, multicenter recruitment with several clinicians, and inclusion of nulliparous participants—a group in whom concern about pain is a known barrier in contraceptive choice making.^{4,30} Pain was measured immediately after placement using a validated scale, minimizing recall bias.

Limitations

The present study was not designed to compare intrauterine mepivacaine with other pain management strategies. Therefore, no conclusions can be drawn regarding relative efficacy vs other pain relief approaches. Limitations include the absence of clinician-assessed placement difficulty and participant anxiety before the procedure, both of which have been associated with higher pain during IUD placement.^{3,31-33} In addition, the exact position of the hydrosalpinx catheter tip was not confirmed with ultrasound. However, absence of ultrasound use during placement adds further to the generalizability to settings where this is not immediately available. The small absolute mean difference (2.0 mm) between the primary analysis and per-protocol analysis suggests that exact tip placement may not be critical for effectiveness.

Conclusions

Intrauterine instillation of 10 mL of mepivacaine (20 mg/mL) significantly reduced pain and improved tolerability during IUD placement compared with instillation of placebo saline.

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